

Windows lab setup — Mastering MongoDB

Technician handout (Word): [Mastering_MongoDB_Lab_Setup_Windows.docx](#).

Set up a Windows laptop **before Day 1**. Every hands-on lab in this course runs in PowerShell against a MongoDB server on `localhost:27017`, with the course folder open so paths such as `.\datasets\training_store\load.js` work.

Course	Mastering MongoDB (3 days)
Organization	Innovation In Software Corporation
OS	Windows 10 or Windows 11, 64-bit
Shell	Windows PowerShell (the labs are written for PowerShell, not Command Prompt or Git Bash)
Time	About 30–45 minutes, plus downloads

Day 1 Lab 1 walks through the same install in class. Completing this page first means that lab is a verification, not a download.

What the laptop needs

Item	Minimum
Windows	10 or 11, 64-bit
Rights	A local account that can install software and start a Windows service (Administrator for <code>Start-Service</code>)
Memory	8 GB RAM
Disk	10 GB free (MongoDB plus the course folder)
Network	Internet for the installers. After that, Days 1–2 labs work offline on the local server. Day 3 Lab 6 needs the instructor’s replica-set or Atlas address, which does need a network.

You do **not** need prior MongoDB experience. You do need to be comfortable opening PowerShell and pasting a command.

Software to install

Install these from MongoDB’s own download pages. Use the current **MongoDB 8.x** Community Edition (the screens in this guide show **8.3.11**). The labs also run on 7.0 or later.

Software	Required?	Used for	Download
MongoDB Community Server	Yes	The database (<code>mongod</code>), installed as a Windows service named <code>MongoDB</code> , listening on <code>localhost:27017</code>	mongodb.com/try/download/community — screens show 8.3.11 , Windows x64, msi
MongoDB Compass	Yes	Opens at the end of the server install. Day 1 Lab 1, Step 10 uses the same saved connection	Installed by the server MSI. Separate download only if that checkbox was cleared: mongodb.com/try/download/compass
MongoDB Shell (mongosh)	Yes	Every lab command	mongodb.com/try/download/shell — screens show 2.12.0 , Windows x64, msi
MongoDB Database Tools	Yes for Day 3 Lab 7	<code>mongodump</code> and <code>mongorestore</code>	mongodb.com/try/download/database-tools — screens show 100.19.0 , Windows x86_64, msi
Course folder	Yes	<code>datasets\training_store\load.js</code> and the lab guides	Git clone or the zip the instructor sends
VS Code or Cursor	Recommended	The lab terminal is PowerShell inside the editor (<code>Ctrl+`</code>)	code.visualstudio.com
Git for Windows	Recommended	Clone the course repository	git-scm.com/download/win
Node.js 18+	Optional	Module 8 connection demo in <code>sample-app/</code>	nodejs.org
Python 3	Optional	The same demo in Python (<code>pip install -r sample-app/requirements.txt</code>)	python.org

Compass is optional inside the Community Server installer. Install it anyway — Lab 1 asks you to open the same document in Compass.

Database Tools are a **separate** installer. They are not included in the server MSI. Without them, Lab 7 Step 4 falls back to a written exercise.

Docker is an optional substitute for the server install. See [Docker instead of the MSI](#) below. You still need `mongosh`, Compass, and Database Tools on the Windows PATH.

1. Install MongoDB Community Server

1. Open [MongoDB Community Server](https://mongodb.com/try/download/community).
2. Choose the current **8.x** (the screen shows **8.3.11**), **Windows x64**, package **msi**, then **Download**.

MongoDB Community Server Download

Community Server

The Community version of our distributed database offers a flexible document data model along with support for:

- Ad-hoc queries
- Secondary indexing
- Real-time aggregations for powerful ways to access and analyze your data

[Select Package](#)

Or Try MongoDB Atlas

Try our fully managed service, MongoDB Atlas, for free with capabilities designed to support you from prototype to production, including:

- Auto-scaling
- High availability by default
- Automatic backups
- Integrated full-text search, vector search, and stream processing
- Multi-region and multi-cloud support

[Get Started For Free](#)

Version

8.3.11 (current) 

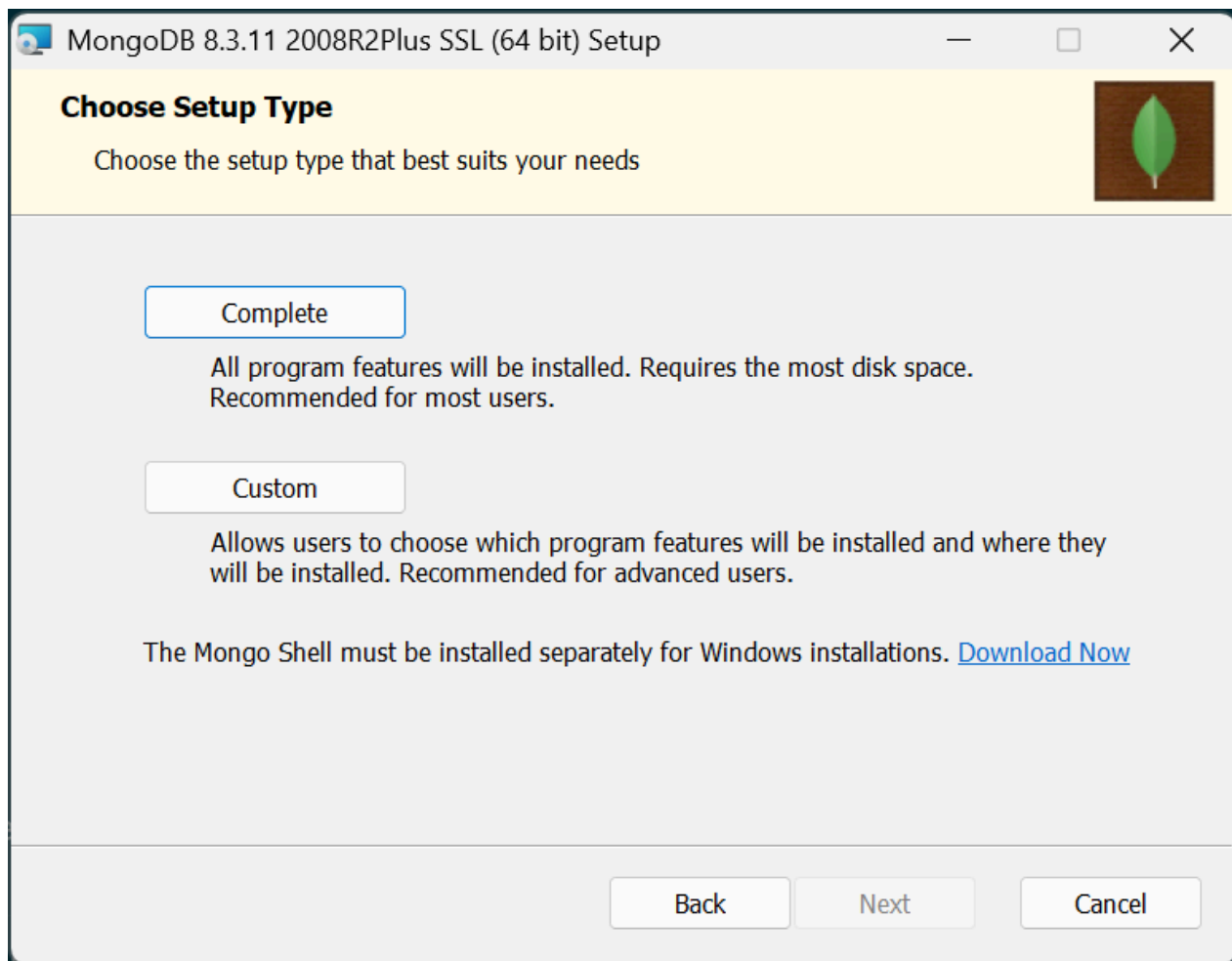
Platform

Windows x64 

Package

msi [Download](#) [Copy link](#)[More Options](#) 

3. Run the MSI and choose **Complete**. The shell is not part of this package.



4. Choose **Install MongoDB as a Service**. Leave **Run service as Network Service user** selected. Do not type a password. - Service name: MongoDB - Data directory: C:\Program Files\MongoDB\Server\8.3\data (accept the path the installer shows) - Log directory: C:\Program Files\MongoDB\Server\8.3\log

MongoDB 8.3.11 2008R2Plus SSL (64 bit) Service Customization

Service Configuration

Specify optional settings to configure MongoDB as a service.

☒ Install MongoDB as a Service

☒ Run service as Network Service user

☐ Run service as a local or domain user:

Account Domain:

Account Name:

Account Password:

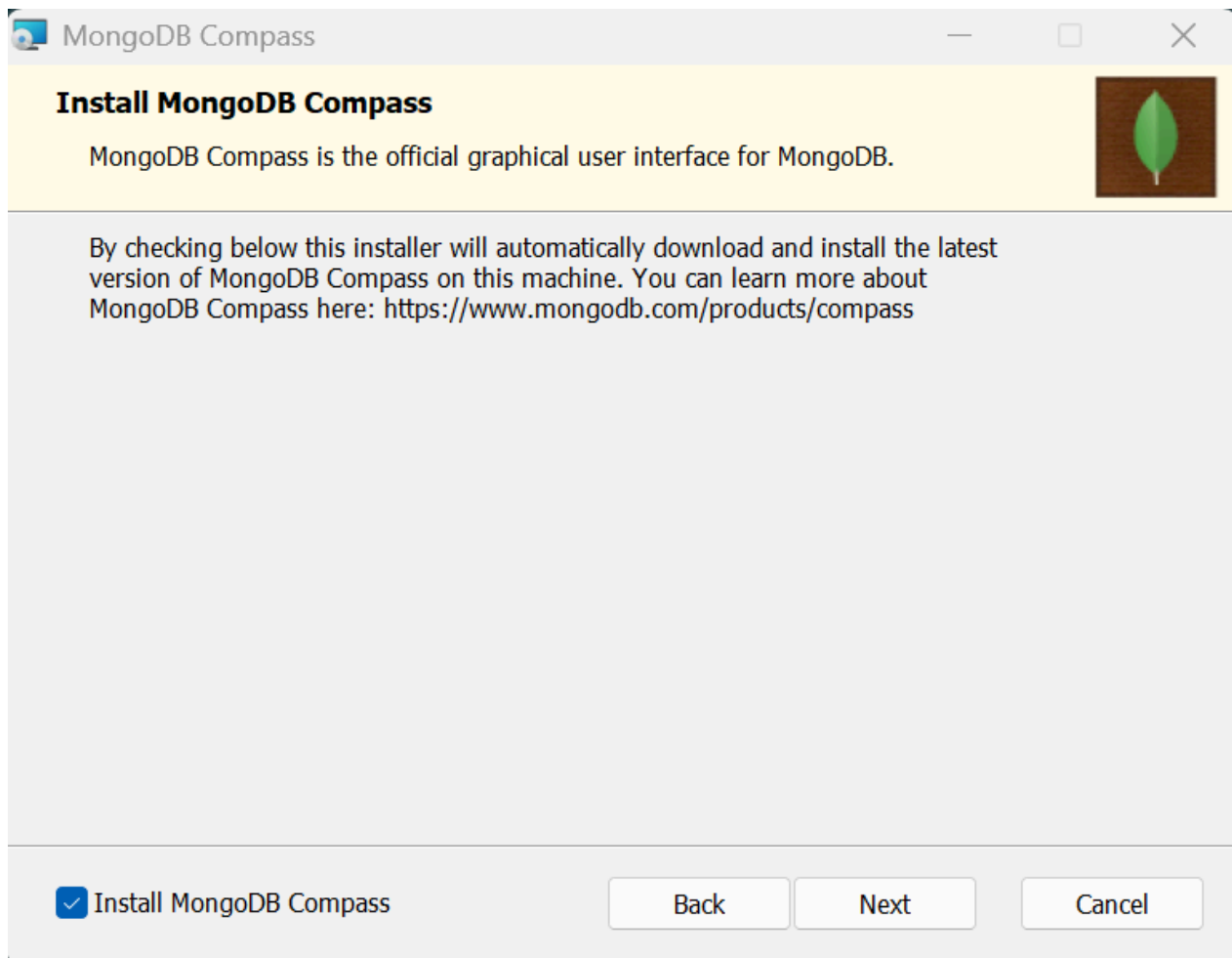
Service Name:

Data Directory:

Log Directory:

< Back Next > Cancel

5. Leave **Install MongoDB Compass** checked.



6. Click **Install** and wait until the wizard finishes. Do not click **Cancel**.

MongoDB 8.3.11 2008R2Plus SSL (64 bit) Setup

Ready to install MongoDB 8.3.11 2008R2Plus SSL (64 bit)

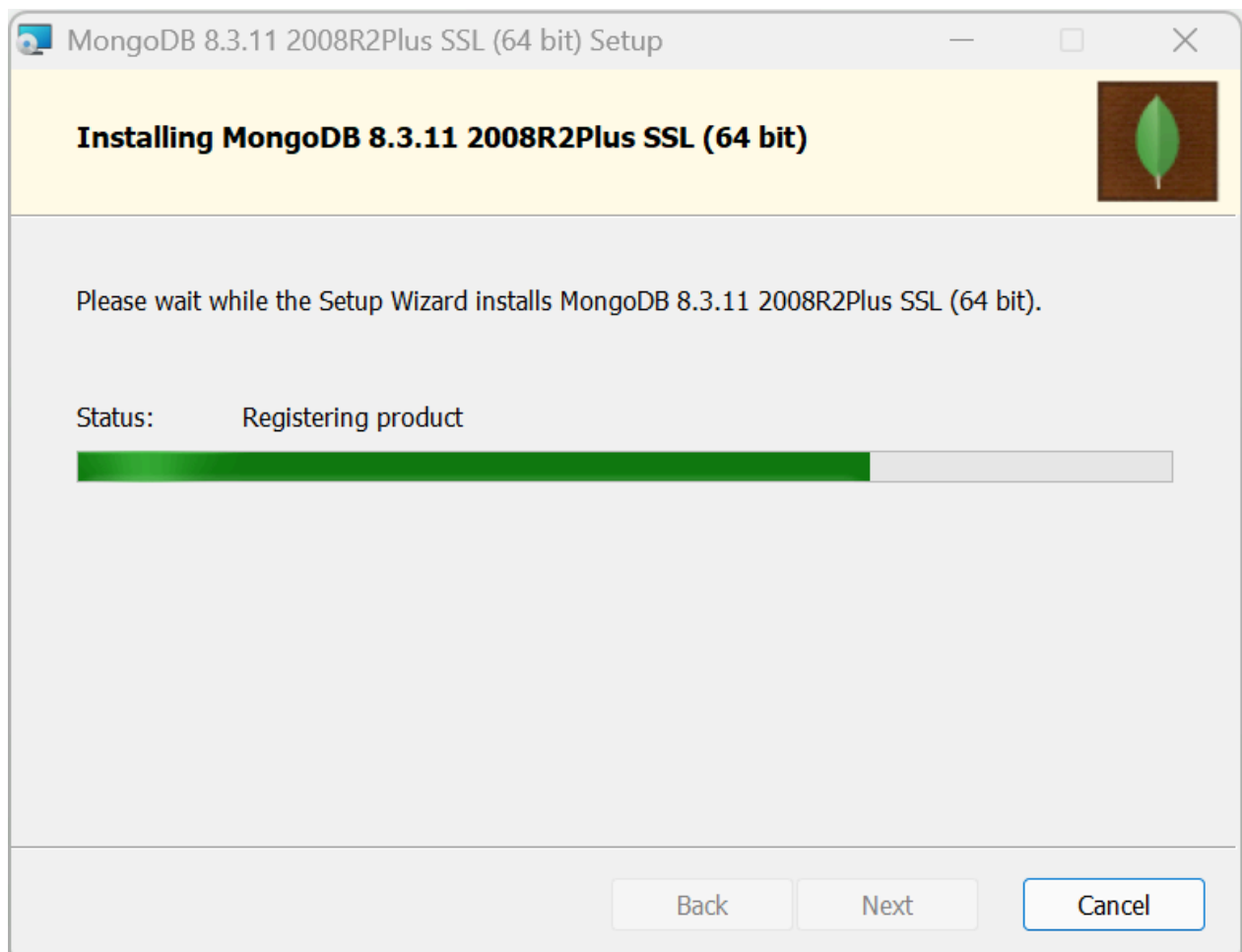


Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.

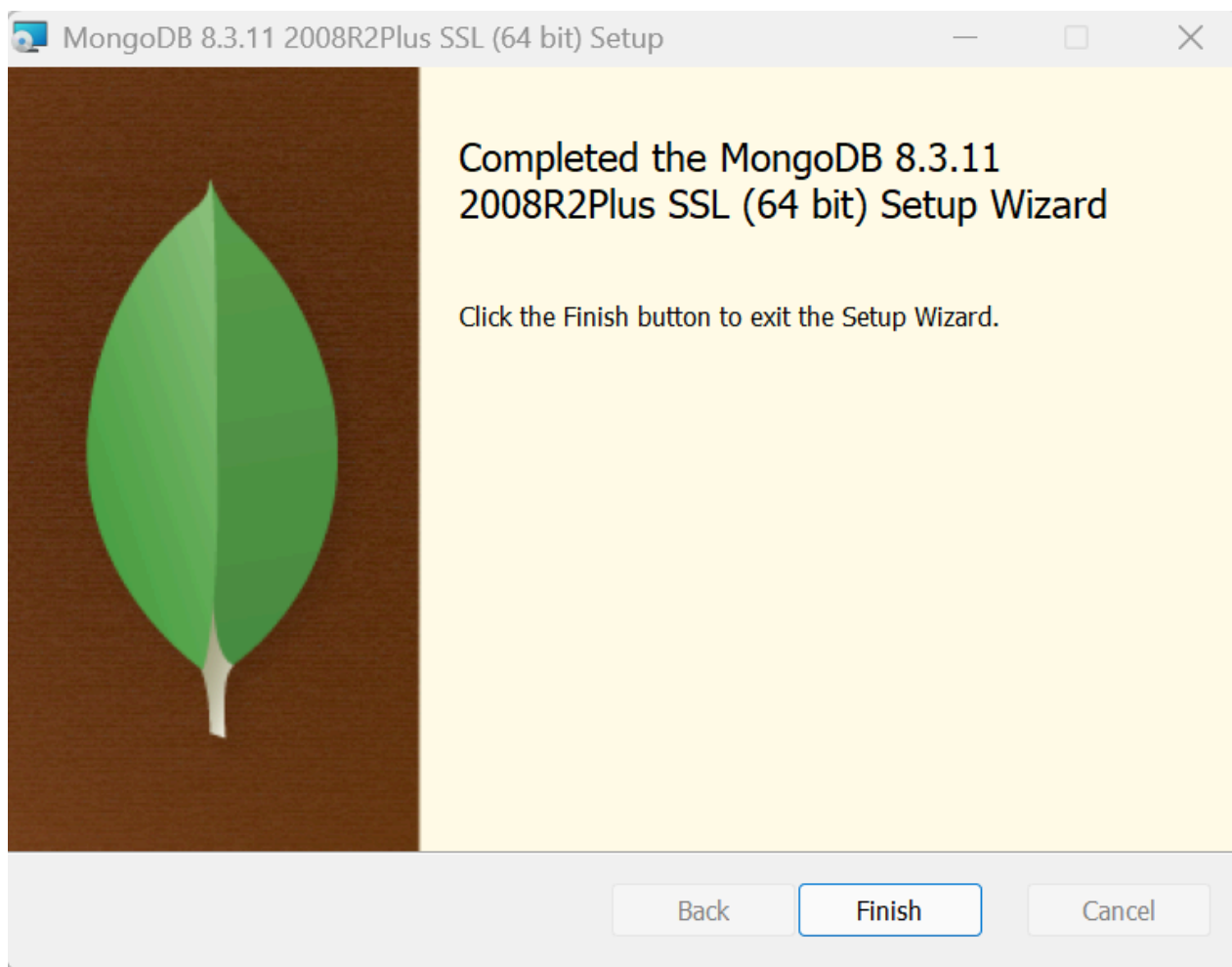
Back

 Install

Cancel



7. Click **Finish**. Do **not** change the server to listen on every network interface.



Files you may need later (the 8.3 folder matches this installer; use the version you actually installed):

File	Path
Server config	C:\Program Files\MongoDB\Server\8.3\bin\mongod.cfg
Server log	C:\Program Files\MongoDB\Server\8.3\log\mongod.log

In `mongod.cfg`, leave these as they are:

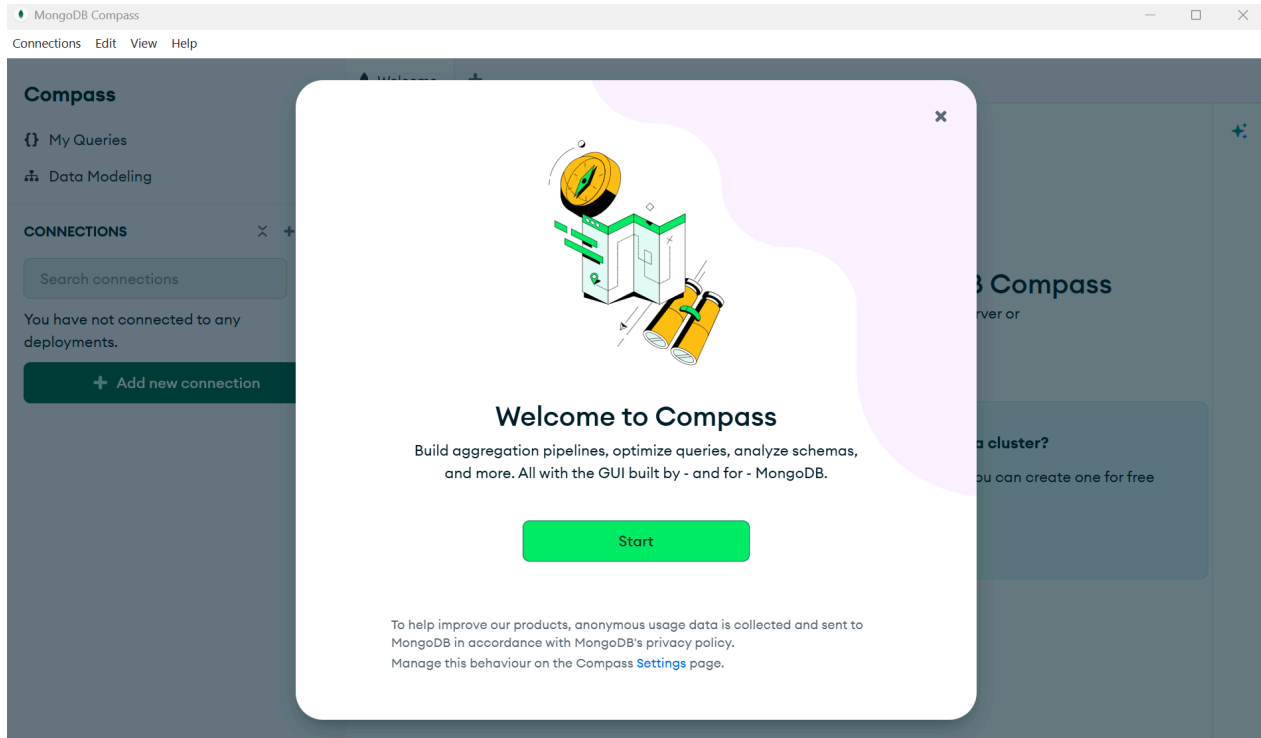
```
net:
  port: 27017
  bindIp: 127.0.0.1
```

`bindIp: 127.0.0.1` keeps the database on this machine only. A local install has access control off by default, which is acceptable only while nothing outside the laptop can reach port 27017. Do not set `bindIp: 0.0.0.0`.

2. Connect MongoDB Compass

The server installer opens Compass when it finishes. Connect it now, before you install the shell. Use the separate Compass download only if the checkbox in section 1 was cleared.

1. If a welcome dialog appears, click **Start**.



2. Click + **Add new connection**. Leave the URI as `mongodb://localhost:27017` (Compass may show a trailing slash). Do not type a username or password. Click **Save & Connect**.

New Connection

×

Manage your connection settings

URI ⓘ

Edit Connection String ☒

mongodb://localhost:27017/

Name

Color

No Color ▼

☐ **Favorite this connection**

Favoriting a connection will pin it to the top of your list of connections

> Advanced Connection Options

How do I find my connection string in Atlas?

If you have an Atlas cluster, go to the Cluster view. Click the 'Connect' button for the cluster to which you wish to connect.

[See example](#)

How do I format my connection string?[See example](#)

Cancel

Save

Connect

Save & Connect

- The sidebar lists the saved connection. A brand-new server shows `admin`, `config`, and `local` only. `training_store` appears after the first lab write.

Compass



My Queries

Data Modeling

CONNECTIONS (1)



▼ localhost:27017

▶ admin

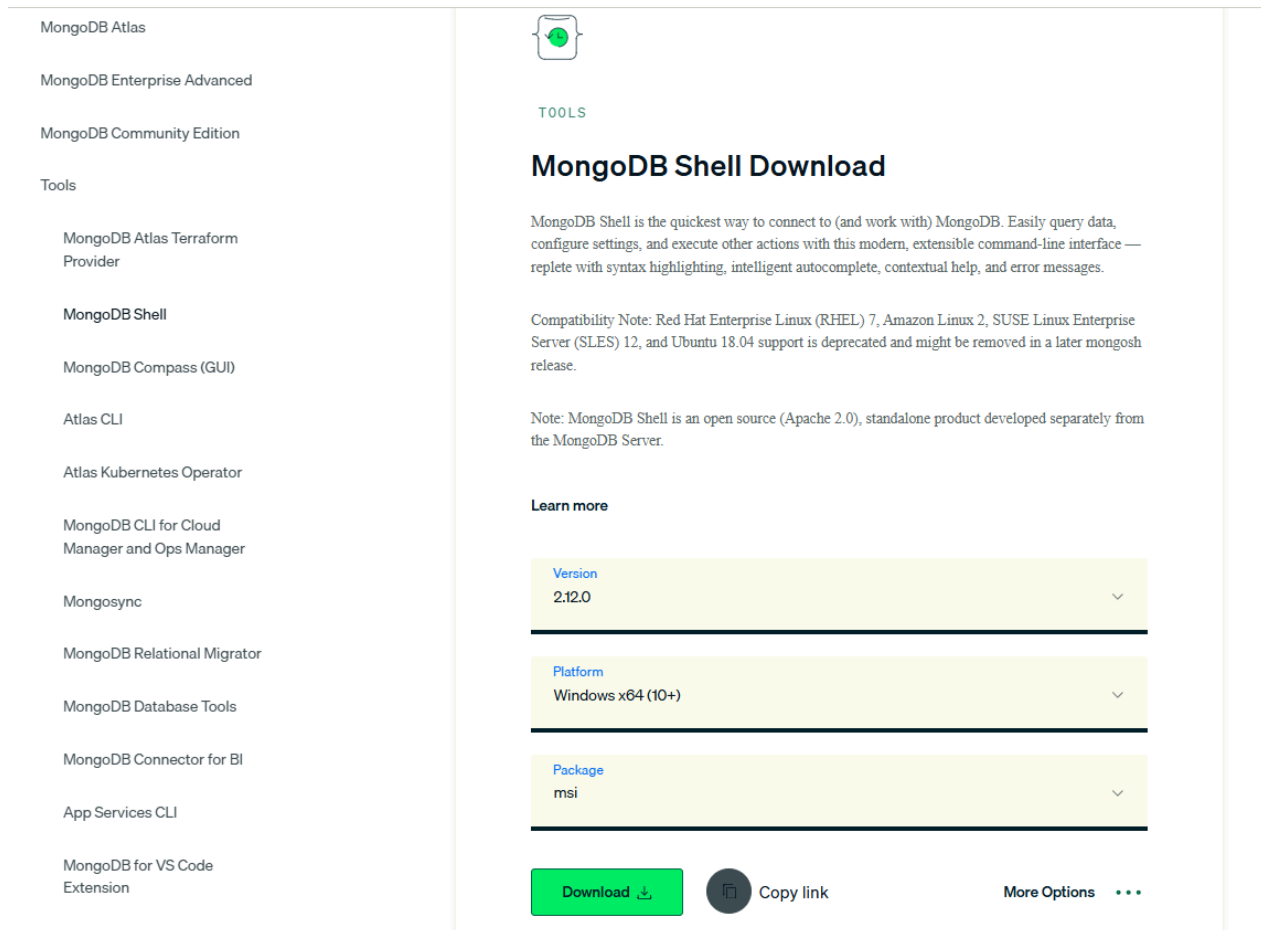
▶ config

▶ local

3. Install `mongosh`

The server MSI does not include `mongosh`. Install the shell from its own MSI (these screens show **2.12.0**). Use the **msi** package. A zip download leaves `mongosh.exe` in a `bin` folder that is not on your `PATH`.

1. Open [MongoDB Shell](#).
2. Choose the current version, **Windows x64 (10+)**, package **msi**, then **Download**.



MongoDB Atlas

MongoDB Enterprise Advanced

MongoDB Community Edition

Tools

- MongoDB Atlas Terraform Provider
- MongoDB Shell**
- MongoDB Compass (GUI)
- Atlas CLI
- Atlas Kubernetes Operator
- MongoDB CLI for Cloud Manager and Ops Manager
- Mongosync
- MongoDB Relational Migrator
- MongoDB Database Tools
- MongoDB Connector for BI
- App Services CLI
- MongoDB for VS Code Extension



TOOLS

MongoDB Shell Download

MongoDB Shell is the quickest way to connect to (and work with) MongoDB. Easily query data, configure settings, and execute other actions with this modern, extensible command-line interface — replete with syntax highlighting, intelligent autocomplete, contextual help, and error messages.

Compatibility Note: Red Hat Enterprise Linux (RHEL) 7, Amazon Linux 2, SUSE Linux Enterprise Server (SLES) 12, and Ubuntu 18.04 support is deprecated and might be removed in a later `mongosh` release.

Note: MongoDB Shell is an open source (Apache 2.0), standalone product developed separately from the MongoDB Server.

Learn more

Version
2.12.0

Platform
Windows x64 (10+)

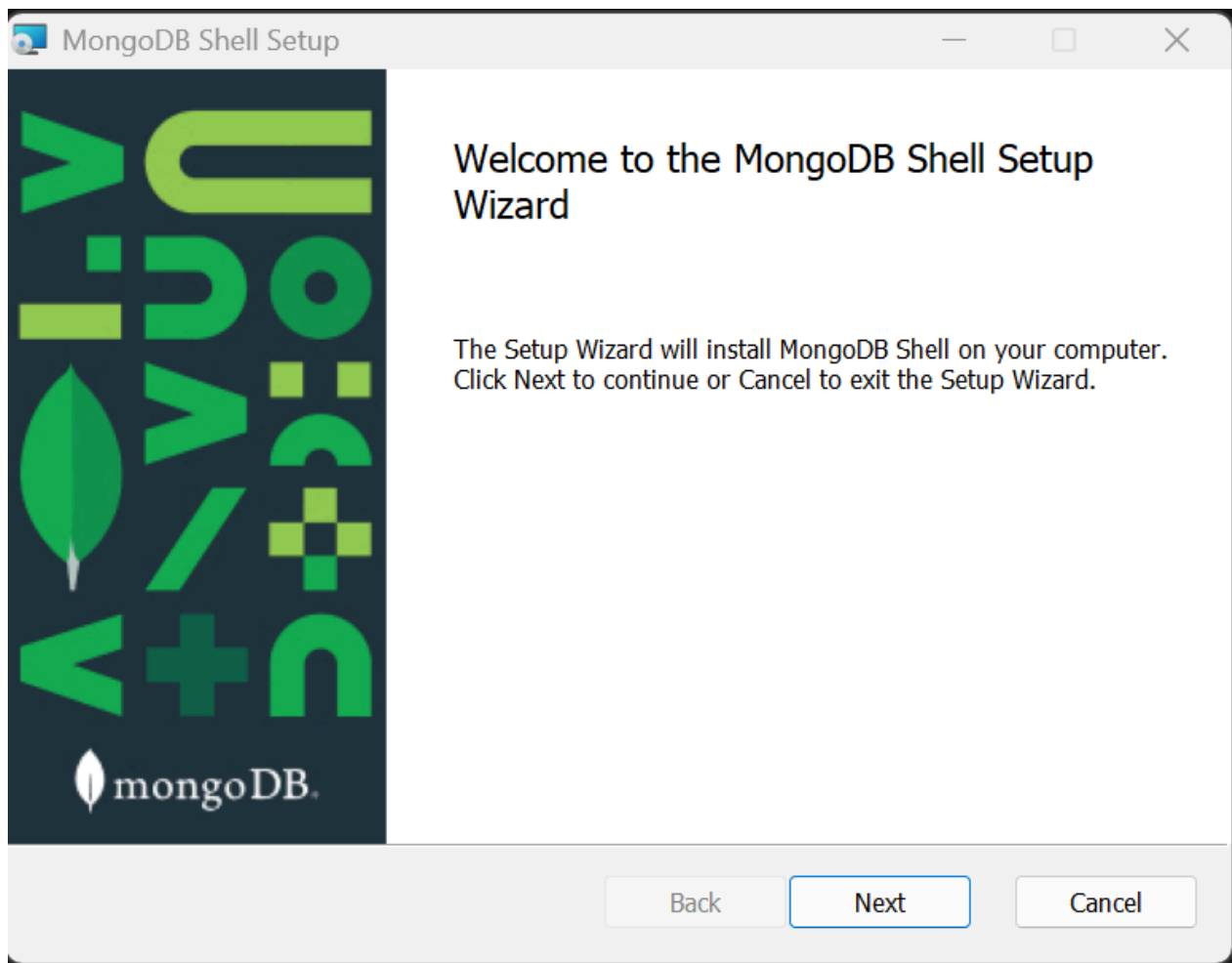
Package
msi

Download 

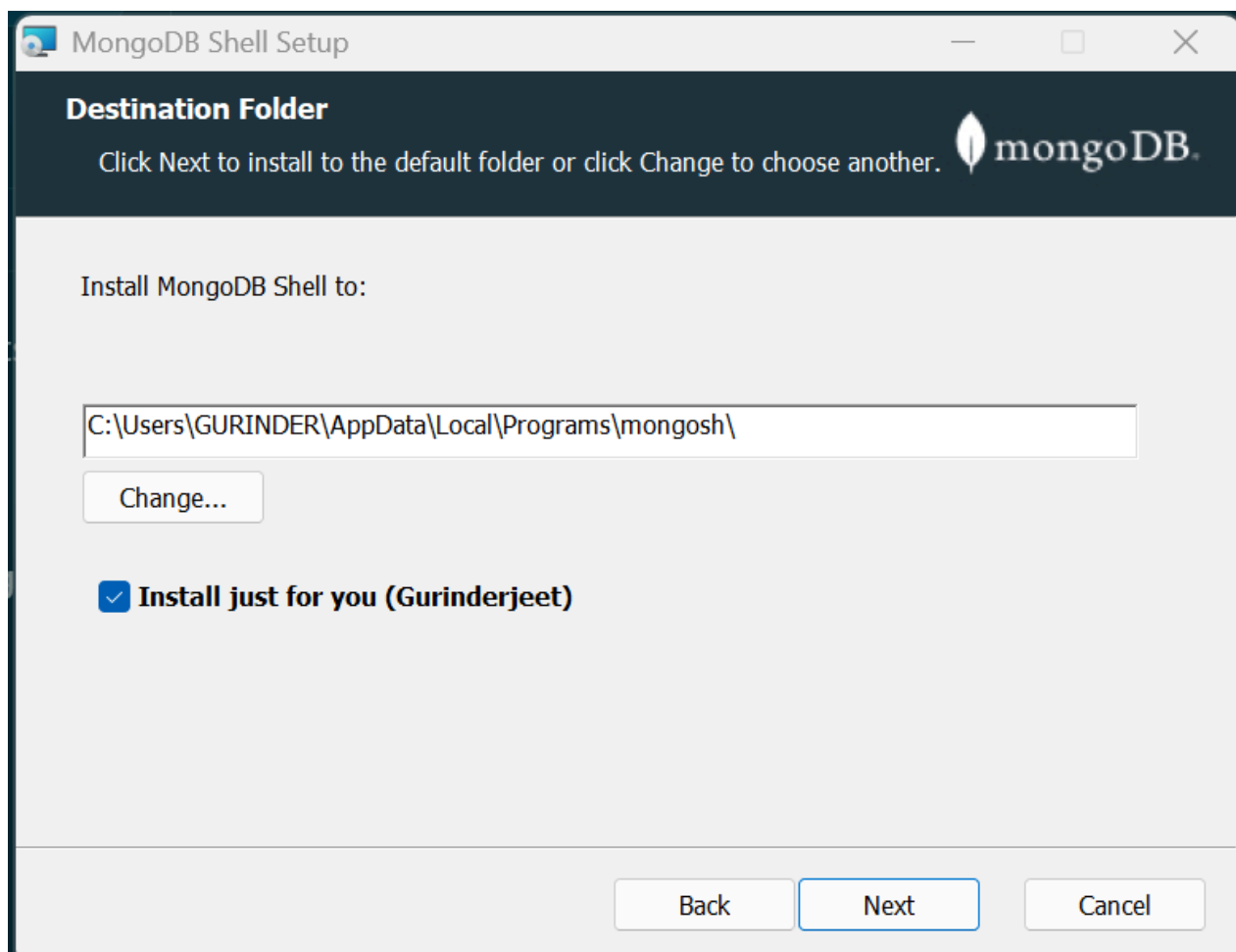
 Copy link

More Options ...

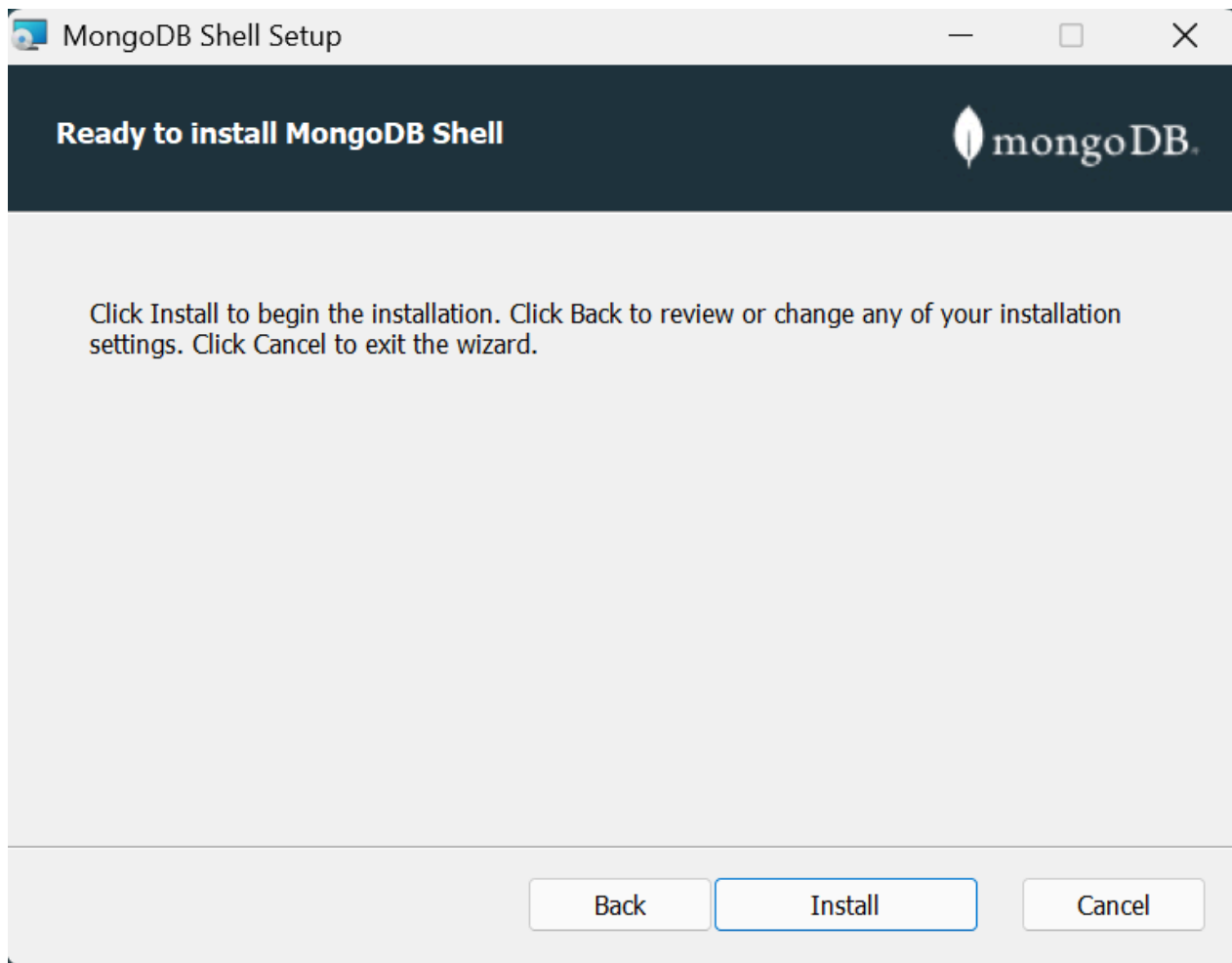
3. Run the MSI. On the welcome page, click **Next**.



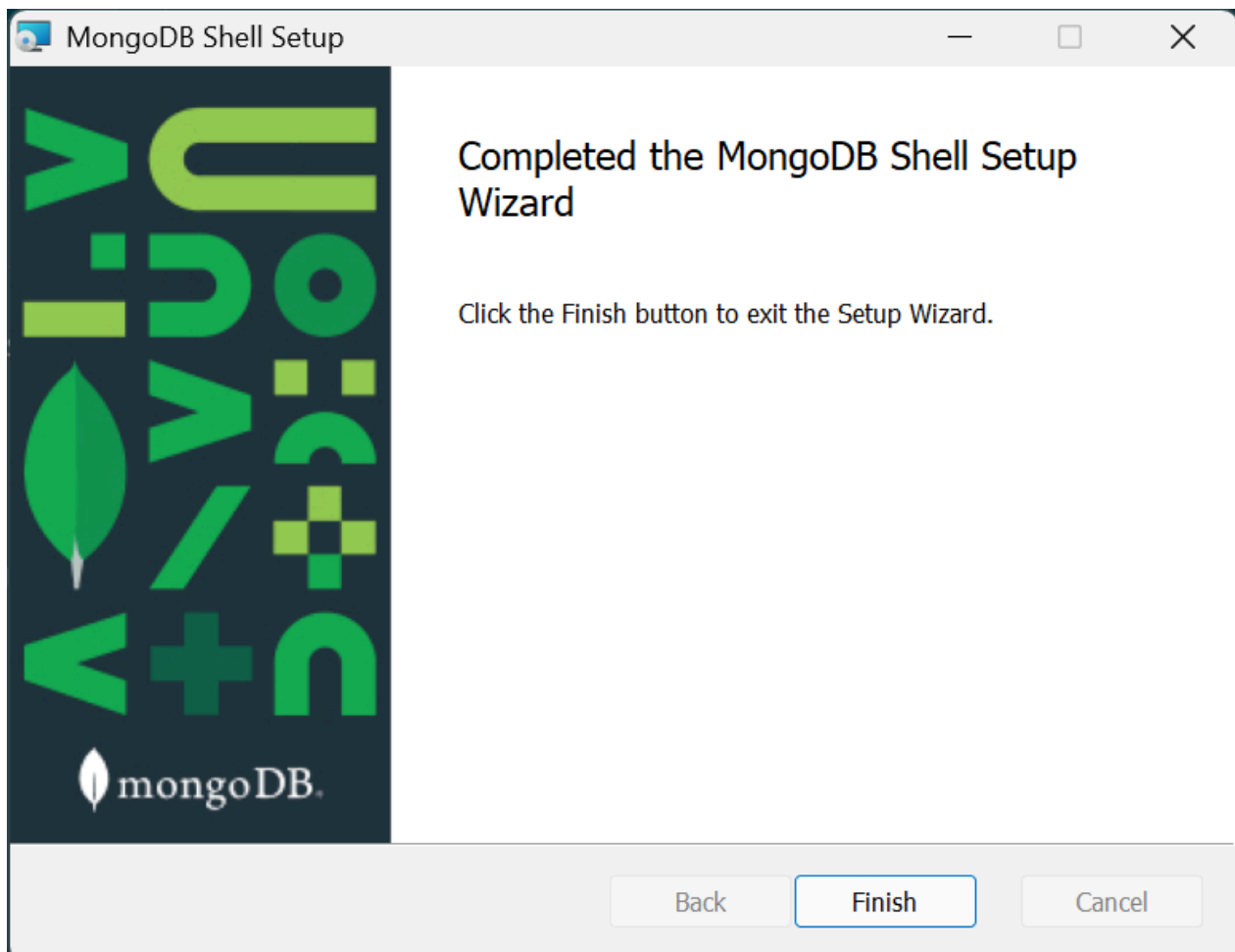
4. If a license page appears, accept it and click **Next**. On **Destination Folder**, leave the folder the wizard shows (under your own user profile) and leave **Install just for you** checked. Click **Next**.



5. Click **Install**.



6. Click **Finish**.



7. Close any open PowerShell windows, including the terminal in VS Code or Cursor, so the new PATH is picked up. Open a new window before you run `mongosh --version`. It should print `2.12.0`.

4. Install MongoDB Database Tools

Database Tools are a separate MSI (these screens show **100.19.0**). The server installer does not include `mongodump` or `mongorestore`, and this MSI does not add them to your PATH.

1. Open [MongoDB Database Tools](#).
2. Choose the current **100.x** version, platform **Windows x86_64**, package **msi**, then **Download**.



TOOLS

MongoDB Command Line Database Tools Download

The MongoDB Database Tools are a collection of command-line utilities for working with a MongoDB deployment. These tools release independently from the MongoDB Server schedule enabling you to receive more frequent updates and leverage new features as soon as they are available. See the [MongoDB Database Tools](#) documentation for more information.

Note: MongoDB Database Tools are distributed as open source (Apache 2.0), standalone products developed separately from the MongoDB Server.

Version

100.19.0



Platform

Windows x86_64



Package

msi



Download

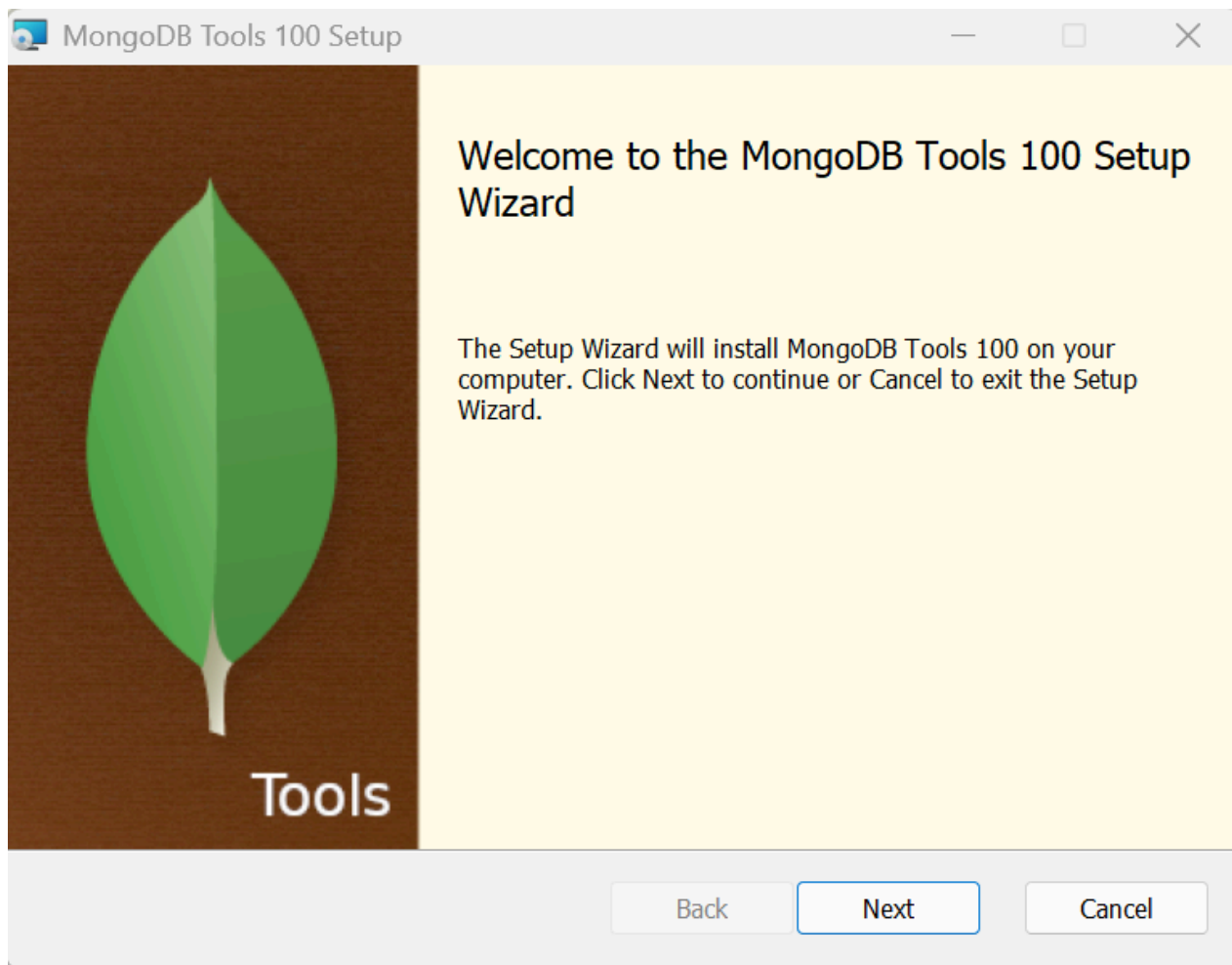


Copy link

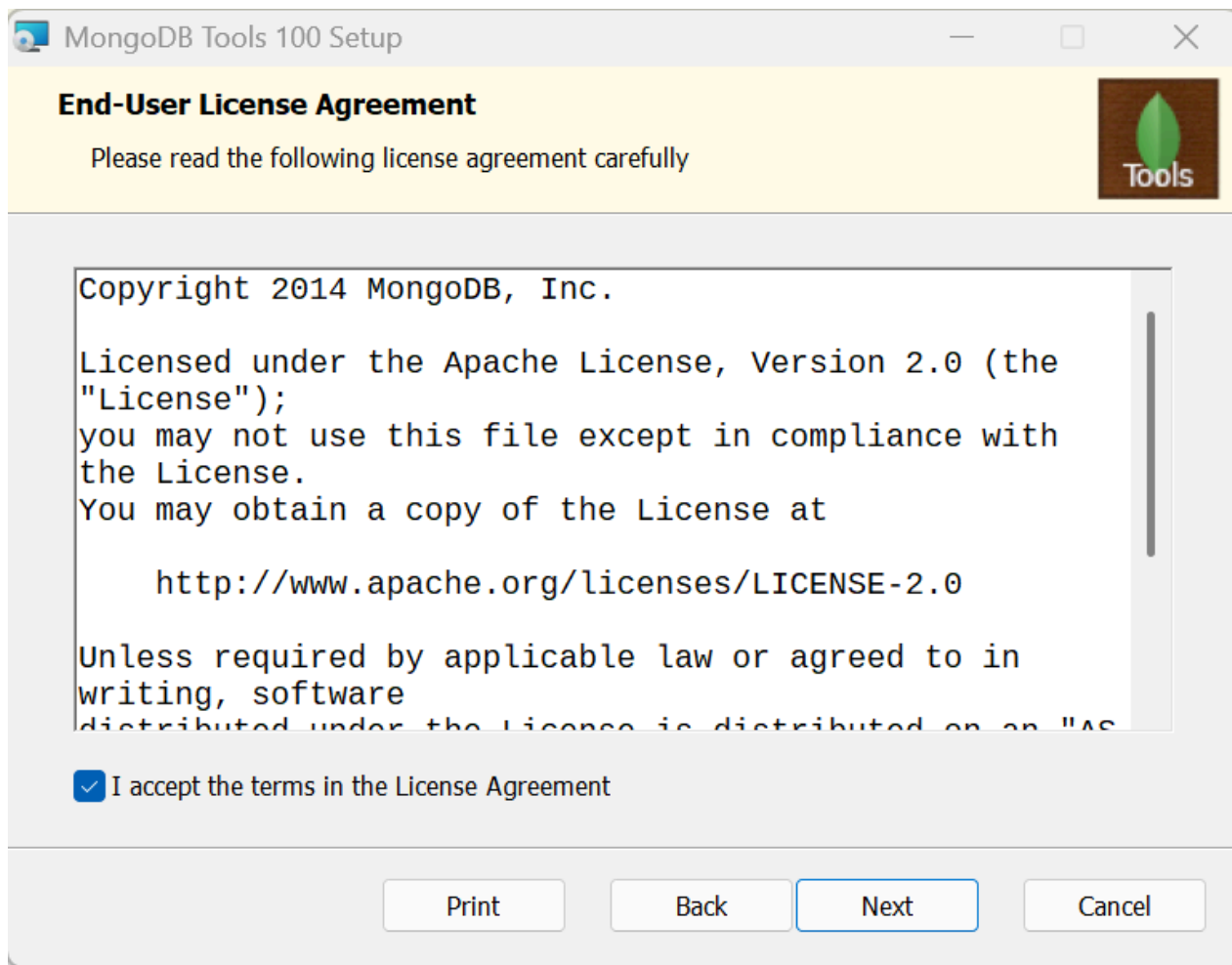
More Options



3. Run the MSI. On the welcome page, click **Next**.



4. Check **I accept the terms in the License Agreement**, then click **Next**.



5. On **Custom Setup**, leave **MongoDB Tools 100** selected and accept the location `C:\Program Files\MongoDB\Tools\100\`. Click **Next**. The wizard installs immediately. Do not click **Cancel**.

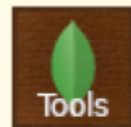


MongoDB Tools 100 Setup

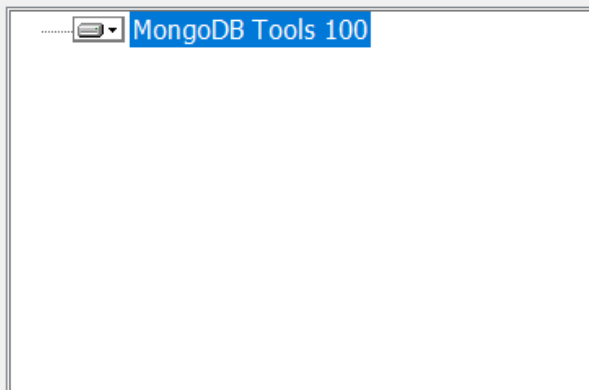


Custom Setup

Select the way you want features to be installed.



Click the icons in the tree below to change the way features will be installed.



MongoDB Tools 100

This feature requires 103MB on your hard drive.

Location: C:\Program Files\MongoDB\Tools\100\

Browse...

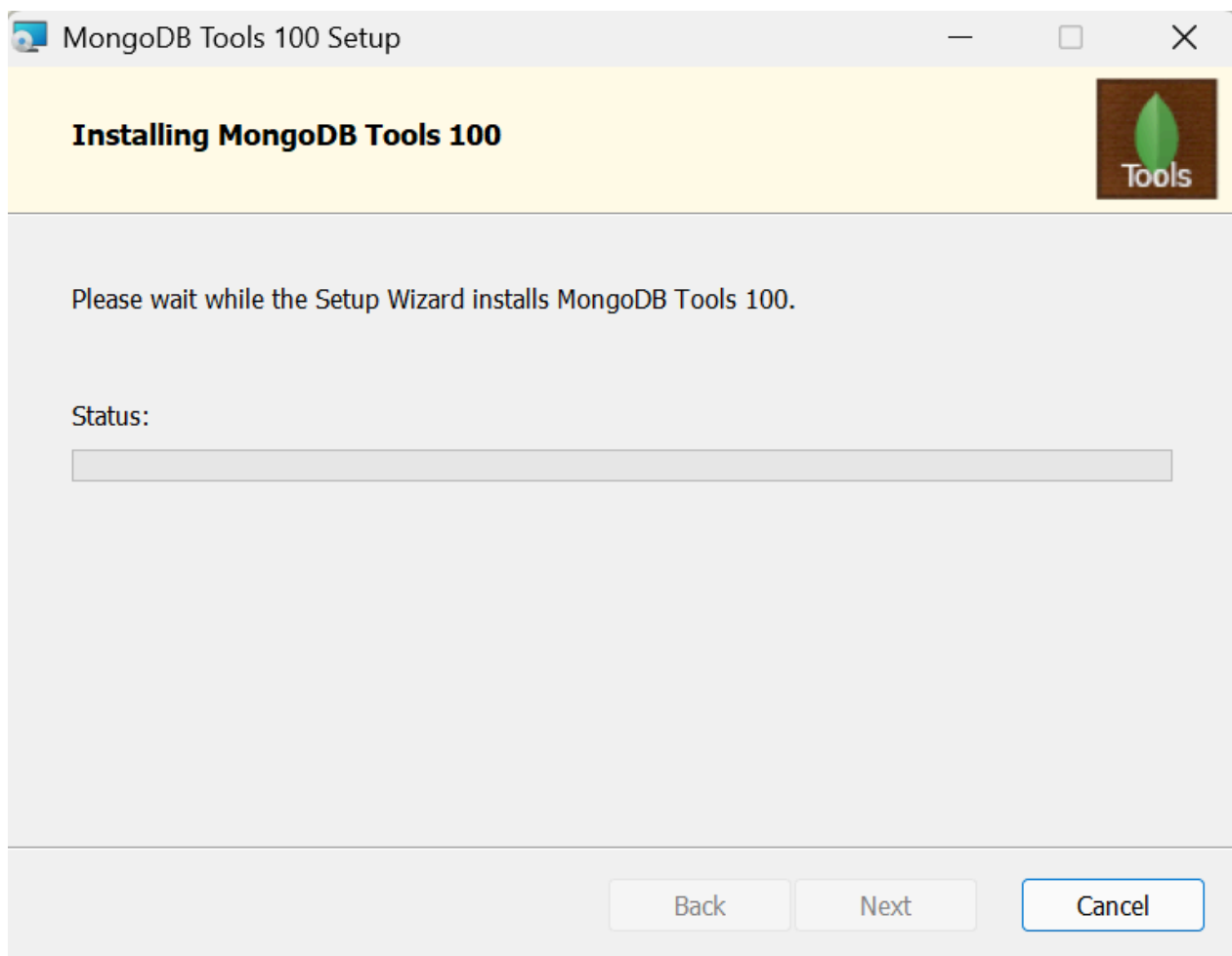
Reset

Disk Usage

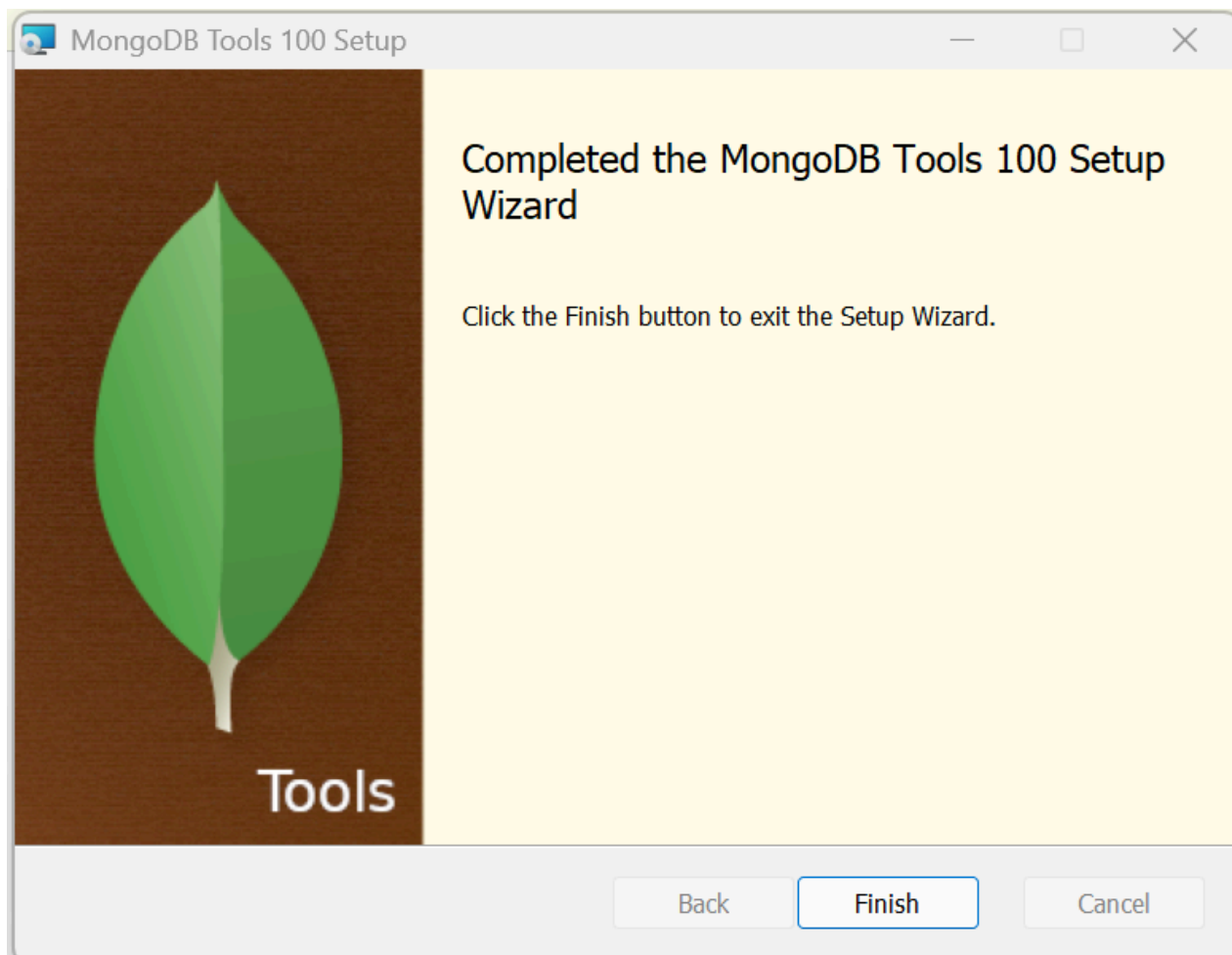
Back

Next

Cancel

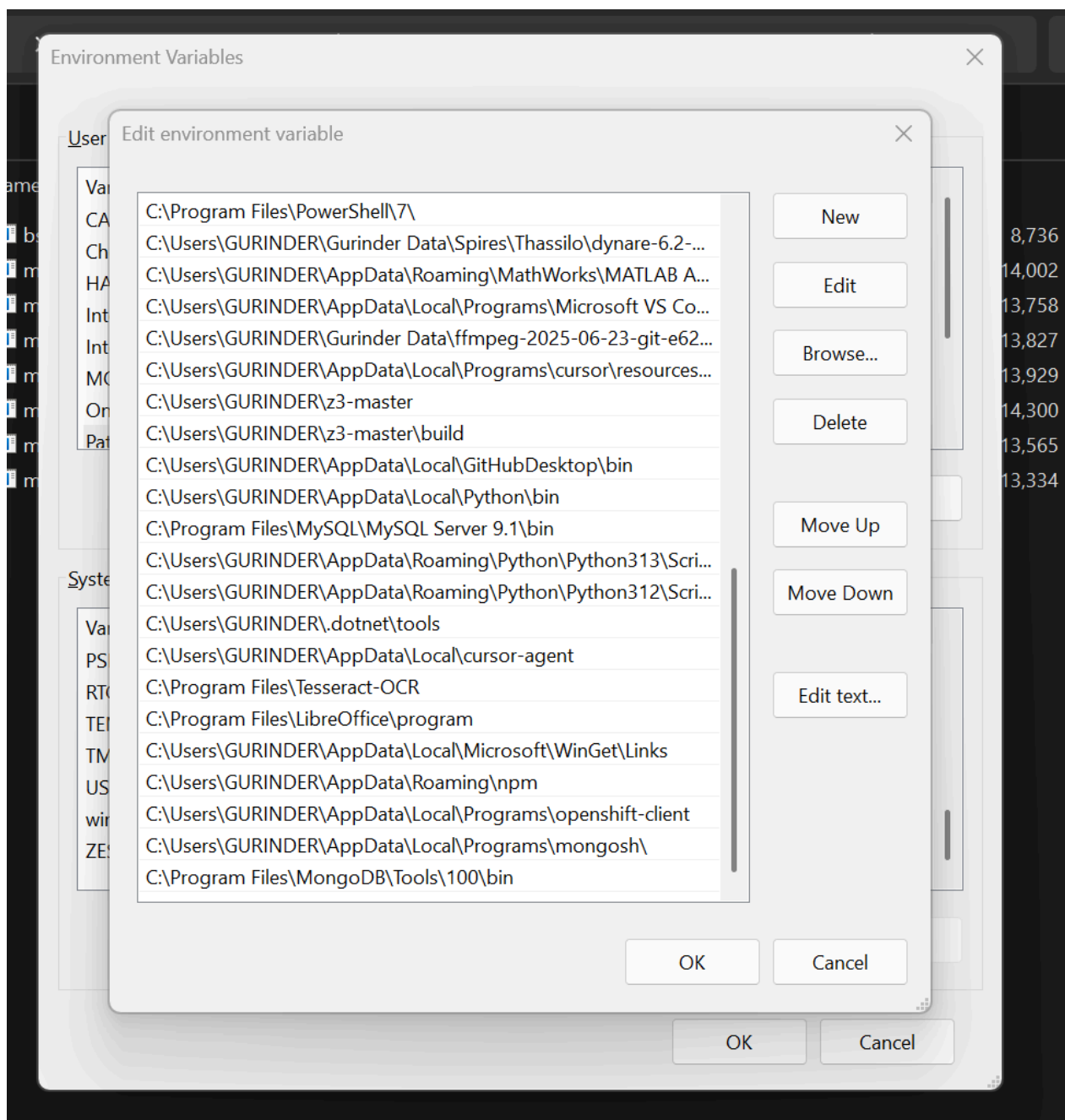


6. Click **Finish**.



7. The commands are in `C:\Program Files\MongoDB\Tools\100\bin`. Add that folder to your user **Path** (the `100` folder matches this installer): - Start → "environment variables" → **Edit environment variables for your account** - Select **Path** → **Edit** → **New** → paste `C:\Program Files\MongoDB\Tools\100\bin` - Click **OK** on the path list, then **OK** on the Environment Variables window

The new entry is the last line. The line above it is the `mongosh` folder from its own installer. Click **OK**. The other lines in that list are this computer's existing folders. Leave them as they are.



8. A PowerShell window that was already open keeps the old Path, so `mongosh`, `mongodump`, and `mongorestore` all fail there even after the installers succeed.


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\GURINDER> mongodump --version
mongodump : The term 'mongodump' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ mongodump --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (mongodump:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\GURINDER> mongorestore --version
mongorestore : The term 'mongorestore' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ mongorestore --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (mongorestore:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\GURINDER> mongosh "mongodb://localhost:27017" --eval "db.runCommand({ ping: 1 })"
mongosh : The term 'mongosh' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ mongosh "mongodb://localhost:27017" --eval "db.runCommand({ ping: 1 } ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (mongosh:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\GURINDER> |
```

Close that window. In a new window, if mongodump is still not recognized, reload Path from the saved settings and try again:

```
$env:Path = [Environment]::GetEnvironmentVariable("Path","Machine") + ";" +
[Environment]::GetEnvironmentVariable("Path","User")
mongosh --version
mongodump --version
mongorestore --version
```

mongosh --version should print 2.12.0. mongodump --version and mongorestore --version should print 100.19.0.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\GURINDER> $env:Path = [Environment]::GetEnvironmentVariable("Path","Machine") + ";" + [Environment]::GetEnvi
ronmentVariable("Path","User")
PS C:\Users\GURINDER> mongosh --version
2.12.0
PS C:\Users\GURINDER> mongodump --version
mongodump version: 100.19.0
git version: a93207alcecc31456ac31678c0406b68c5c929aa
Go version: go1.26.8
os: windows
arch: amd64
compiler: gc
PS C:\Users\GURINDER> mongorestore --version
mongorestore version: 100.19.0
git version: a93207alcecc31456ac31678c0406b68c5c929aa
Go version: go1.26.8
os: windows
arch: amd64
compiler: gc
PS C:\Users\GURINDER>
```

An Administrator window can see this Path before a normal window does, because elevation reads the saved Path fresh. Sign out of Windows and back in so every new normal window sees it without the reload line.

5. Get the course folder

Labs load data with a path relative to the repository root. Open that folder in VS Code or Cursor; do not run the commands from `C:\Users\<you>`.

With Git:

```
git clone <repository-url>
cd MasteringMongoDB
```

Or unzip the folder the instructor sends, then `cd` into `MasteringMongoDB`.

You should see `datasets\training_store\load.js` and `labs\`.

6. Confirm the machine is ready

Open a **new** PowerShell window (so PATH changes apply). Run these in order.

Service is running

```
Get-Service MongoDB
```

Expected: `Status : Running`.

```

PS C:\Users\GURINDER> Get-Service MongoDB

Status      Name      DisplayName
-----
Running     MongoDB   MongoDB Server (MongoDB)

PS C:\Users\GURINDER> Test-NetConnection localhost -Port 27017
WARNING: TCP connect to (::1 : 27017) failed

ComputerName      : localhost
RemoteAddress     : 127.0.0.1
RemotePort        : 27017
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True

PS C:\Users\GURINDER> mongosh "mongodb://localhost:27017"

```

If it says `Stopped`, open PowerShell **as Administrator** and run:

```
Start-Service MongoDB
```

Port 27017 answers on this machine

```
Test-NetConnection localhost -Port 27017
```

Expected: `TcpTestSucceeded : True`.

PowerShell may also print `WARNING: TCP connect to (::1 : 27017) failed`. That line is IPv6. This server listens on `127.0.0.1` only, so the IPv4 row is the one that matters: `RemoteAddress : 127.0.0.1` and `TcpTestSucceeded : True`. The warning does not mean the install failed.

Clients are on PATH

```

mongosh --version
mongodump --version
mongoexport --version

```

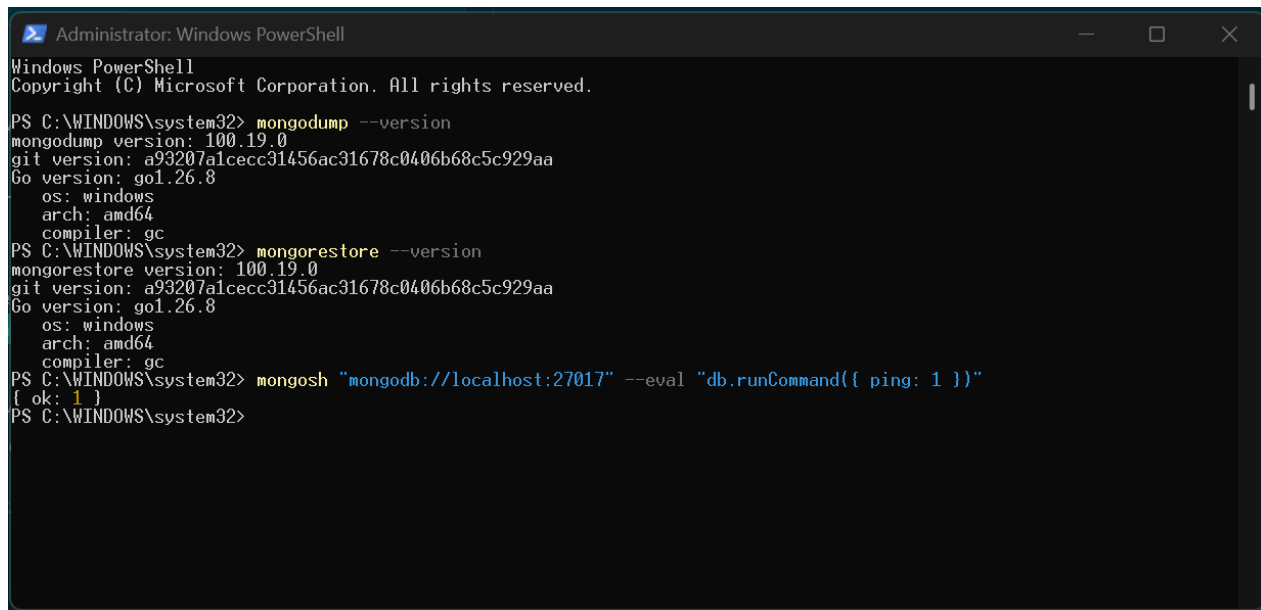
Each command prints a version. On this walkthrough that is `mongosh 2.12.0` and Database Tools `100.19.0`. `mongosh is not recognized` or `mongodump is not recognized` means that installer is missing or the terminal was opened before PATH was updated.

Shell can talk to the server

```
mongosh "mongodb://localhost:27017" --eval "db.runCommand({ ping: 1 })"
```

Expected: a document with `ok: 1`. Then type `exit` if a prompt is still open.

These three commands work from any folder. An Administrator window is not required.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> mongodump --version
mongodump version: 100.19.0
git version: a93207a1cecc31456ac31678c0406b68c5c929aa
Go version: go1.26.8
os: windows
arch: amd64
compiler: gc
PS C:\WINDOWS\system32> mongorestore --version
mongorestore version: 100.19.0
git version: a93207a1cecc31456ac31678c0406b68c5c929aa
Go version: go1.26.8
os: windows
arch: amd64
compiler: gc
PS C:\WINDOWS\system32> mongosh "mongodb://localhost:27017" --eval "db.runCommand({ ping: 1 })"
{ ok: 1 }
PS C:\WINDOWS\system32>
```

Compass

Start Compass from the Start menu. + **Add new connection** → `mongodb://localhost:27017` → **Save & Connect**. You should see the server. A brand-new server lists `admin`, `config`, and `local` only. `training_store` appears after the first lab write.

Readiness checklist

Tick these before class. This is the same list as [Exercise 2.4](#), filled in early.

- ☐ Windows 10 or 11, and you can open PowerShell
- ☐ MongoDB service MongoDB is **Running**
- ☐ `Test-NetConnection localhost -Port 27017` shows `TcpTestSucceeded : True`
- ☐ `mongosh --version` prints a version
- ☐ `mongosh "mongodb://localhost:27017"` reaches a prompt, and `db.runCommand({ ping: 1 })` returns `ok: 1`
- ☐ Compass opens and connects to `mongodb://localhost:27017`
- ☐ `mongodump --version` and `mongorestore --version` both print a version
- ☐ The course folder is on disk and contains `datasets\training_store\load.js`

- [] `mongod.cfg` still has `bindIp: 127.0.0.1` and port `27017`
- [] No password or full Atlas URI is saved in a file you might commit or share

Class then continues with [Lab 1 — Install, Connect and Verify](#).

How each lab uses this setup

When	What must already work
Day 1 Lab 1	Server service, <code>mongosh</code> , Compass
Day 1 Lab 2 through Day 3 Lab 5	Same local server, plus the course folder so <code>load.js</code> reloads <code>training_store</code>
Day 3 Lab 6	The instructor's replica-set or Atlas URI. A standalone <code>mongod</code> on <code>localhost</code> is not a replica set.
Day 3 Lab 7	<code>mongodump</code> and <code>mongorestore</code> on PATH, local server on <code>localhost:27017</code>
Module 8 <code>sample-app</code> (optional)	Node.js 18+ or Python 3, and <code>\$env:MONGODB_URI = "mongodb://localhost:27017"</code>

Reload the sample data from the repository root whenever a lab tells you to:

```
mongosh "mongodb://localhost:27017" .\datasets\training_store\load.js
```

That script drops and reloads `training_store`. Run it from PowerShell at the repo root, not from inside an already-open `mongosh` prompt.

Docker instead of the MSI

Use this only if you cannot install the Windows service. You still install `mongosh`, Compass, and Database Tools on Windows.

```
docker run -d --name mongo `
  -p 127.0.0.1:27017:27017 `
  -v mongo-data:/data/db `
  mongodb/mongodb-community-server
```

`-p 127.0.0.1:27017:27017` publishes the port on this machine only. The named volume keeps data across container restarts. Connect the same way: `mongodb://localhost:27017`. There is no MongoDB Windows service in this path; `docker ps` should show `mongo` running.

Atlas instead of a local server

Use Atlas only when the instructor assigns it. Days 1–2 can run against Atlas. Lab 7's dump and restore commands in the guide target `localhost:27017`; a local server is the path those steps assume.

1. Sign in to the Atlas project the instructor names.
2. Create a database user with `readWrite` on `training_store`. Do not reuse your Atlas login as the database user.
3. Under **Network Access**, add **your current public IP**. Do not add `0.0.0.0/0`.
4. Store the `mongodb+srv://` host in an environment variable **without** the password:

```
$env:MONGODB_URI = "mongodb+srv://<cluster-host>/"  
mongosh $env:MONGODB_URI --username <username>
```

`mongosh` prompts for the password. Do not put the password in the command, in a slide, or in chat.

Security rules for the whole course

- Placeholders only in notes and chat: `<connection-string>`, `<username>`, `<password>`.
 - Let `mongosh` and Compass prompt for the password.
 - Keep the local server on `127.0.0.1`. Do not open port 27017 on the Windows firewall for other machines.
 - Do not enable authentication or create users on a shared classroom instance unless the instructor says to.
-

If something fails

Change one thing, then retry `mongosh "mongodb://localhost:27017" --eval "db.runCommand({ ping: 1 })"`.

What you see	What to do
Get-Service MongoDB → Stopped	Administrator PowerShell: Start-Service MongoDB
Start-Service says access denied	The window is not elevated. Close it and choose Run as administrator .
ECONNREFUSED on localhost:27017	The service is stopped, or another program owns port 27017. Check the service, then Test-NetConnection localhost -Port 27017.
Service will not start	Open C:\Program Files\MongoDB\Server\8.3\log\mongod.log (use your installed version). Typical causes: the data directory is missing or not writable, or port 27017 is already taken.
mongosh, mongodump, and mongorestore are all not recognized in a window that was already open	That window kept the Path from before the install. Close it. In a new window run \$env:Path = [Environment]::GetEnvironmentVariable("Path","Machine") + ";" + [Environment]::GetEnvironmentVariable("Path","User"), then the version commands. Sign out and back in so later windows do not need that line.
mongosh is not recognized in a new window	Install the shell MSI (the msi , not the zip), then open a new PowerShell window.
mongodump is not recognized in a new window, but an Administrator window works	The tools bin folder is saved, and this sign-in has not reloaded it. Run the Path reload line above, then sign out and back in.
Compass connects but shows no new documents	Use the same URI as mongosh (mongodb://localhost:27017) and refresh the collection.
mongosh works in one terminal and not another	The second terminal was opened before install. Close it and open a new one.

Full connection-failure table used in class: [Exercise 2.3](#).

Related pages

- [Lab 1 — Install, Connect and Verify](#)
- [Exercise 2.4 — Environment Readiness Checklist](#)
- [Sample dataset load](#)
- [Application connection demo](#) (optional, Module 8)